Course: MCA **Semester:** 2

Prerequisite: Knowledge of programming language (C/C++).

Course Objective: To provide understanding of basics of data structures and their operation for efficient storage and retrieval of data.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/	Credit	T	CE	P	T	P	
4	0	2	-	5	20	20	20	60	30	150

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Introduction** Primitive and non-primitive data structures, String manipulation and pattern matching, Storage representation of strings, Text handling, Key Word In Context (KWIC) indexing, Arrays, Storage structure for arrays, Special types of arrays – triangular and sparse.	10	7
2	**Stack and Queue** Stack, Stack operations, Applications of stack - recursion, polish notations - prefix, infix, postfix, Algorithms of stack applications, Introduction to queue, Algorithms and implementation of simple queue, Circular queue, Double ended queue, Priority queue.	20	12
3	**Linked List** Linked list, Algorithms and implementation of singly linked list, Doubly linked list, Circular linked list, Operations on linked list, Applications – polynomial representation, addition of two polynomials.	20	8
4	**Trees** Concept and terminologies of tree, General tree, Binary tree and its storage representation, Binary search tree and its operations – create, insert, delete, Traversal of tree - inorder, preorder, postorder, Threaded tree, B tree and B+ tree, Height balanced tree - AVL tree, Rotations in AVL tree, Applications – heap tree, expression tree.	20	12
5	**Graph** Concept and terminologies of graph, Representation of graph -adjacency matrix, adjacency lists, Introduction to graph traversal - Depth First Search (DFS), Breadth First Search (BFS), Introduction to spanning tree.	10	5
6	**Searching, Sorting and Hashing** Linear search, Binary search, Bubble sort, Selection sort, Insertion sort, Shell sort, Quick sort, Heap sort, Merge sort, Radix sort, Hashing, Hashing functions, Collision resolution techniques.	20	13

Reference Books

1.	**An Introduction to Data Structures with Applications (TextBook)** By Jean-Paul Tremblay, Paul G. Sorenson | Tata McGraw-Hill | 2nd Edition, (2007)
2.	**Introduction to Algorithm** By Cormen, Leiserson, Rivest, Stein | PHI (2003) | 2nd Edition
3.	**Data Structures using C and C++** By Tanenbaum | PHI
4.	**Expert Data Structures with C** By R. B. Patel
5.	**Theory and Problems of Data Structures** By Seymour Lipschutz | Schaum's Outline Series
6.	**Data Structures Through C++** By Yashavant Kanetkar | BPB

Course Outcome
After Learning the Course the students shall be able to:
1. describe the significance of various linear and non-linear data structures such as arrays, stack, queue, linked list, trees and graph. 2. identify the appropriate data structure for a given problem. 3. construct most suitable data structure to solve a problem by considering various problem characteristics such as data size and various type of operations. 4. design and implement various techniques for searching, sorting and hashing.

List of Practical

1.	**Write a program to perform various stack operations using array**
2.	**Write a program to convert infix expression to prefix and postfix expression using stack**
3.	**Write a program to perform insert and remove operations on following** a. Simple Queue b. Circular Queue c. Priority Queue
4.	**Write a program to perform Double Ended Queue [Input Restricted / Output Restricted]**
5.	**Write a program to create a singly link list in FIFO & LIFO form**
6.	**Write a program to perform following singly link list operations** a. insert b. delete c. search d. reverse
7.	**Write a program to create a doubly link list in FIFO & LIFO form**
8.	**Write a program to perform following doubly link list operations** a. insert b. delete c. search d. reverse
9.	**Write a program to add two polynomials**
10.	**Write a program to perform following circular link list operations** a. insert b. delete
11.	**Write a program to create a binary search tree and print its element in** a. Inorder b. Preorder c. Postorder
12.	**Write a program for insertion of a node in B tree / B+ tree**
13.	**Write a program to create a graph in a adjacency list structure traverse it in** a. DFS b. BFS
14.	**Write a program to perform following sort**

	a. Bubble Sort b. Selection Sort c. Insertion Sort d. Shell Sort e. Quick Sort f. Heap Sort g. Merge Sort h. Radix Sort
15.	**Write a program to search an element using** a. Linear Search b. Binary Search